

1. Online Course Registration System

```
package JDBCASSES;
```

```
import java.sql.Connection;
```

```
import java.sql.DriverManager;
```

```
import java.sql.PreparedStatement;
```

```
import java.sql.ResultSet;
```

```
import java.sql.SQLException;
```

```
import java.sql.Statement;
```

```
import java.util.Scanner;
```

```
public class CourseJDBCApp {
```

```
    private static final String URL =  
    "jdbc:mysql://localhost:3306/testdb";
```

```
    private static final String USER = "root";
```

```
    private static final String PASSWORD = "root";
```

```
    public static void main(String[] args) {
```

```
        Scanner scanner = new Scanner(System.in);
```

```
        try (Connection conn = DriverManager.getConnection(URL,  
USER, PASSWORD)) {
```

```
            while (true) {
```

```
                System.out.println("\n--- Course Registration System ---");
```

```

        System.out.println("1. Add Course");
        System.out.println("2. View Courses");
        System.out.println("3. Update Course");
        System.out.println("4. Delete Course");
        System.out.println("5. Exit");
        System.out.print("Choose option: ");
        int choice = scanner.nextInt();

        switch (choice) {
            case 1 -> addCourse(conn, scanner);
            case 2 -> viewCourses(conn);
            case 3 -> updateCourse(conn, scanner);
            case 4 -> deleteCourse(conn, scanner);
            case 5 -> {
                System.out.println("Exiting...");
                return;
            }
            default -> System.out.println("Invalid option.");
        }
    }
} catch (SQLException e) {
    e.printStackTrace();
}
}

```

```
private static void addCourse(Connection conn, Scanner sc)
throws SQLException {
    System.out.print("Enter Course ID: ");
    int id = sc.nextInt();
    sc.nextLine();
    System.out.print("Enter Course Name: ");
    String name = sc.nextLine();
    System.out.print("Enter Faculty: ");
    String faculty = sc.nextLine();
    System.out.print("Enter Credits: ");
    int credits = sc.nextInt();

    String query = "INSERT INTO courses VALUES (?, ?, ?, ?)";
    try (PreparedStatement ps = conn.prepareStatement(query)) {
        ps.setInt(1, id);
        ps.setString(2, name);
        ps.setString(3, faculty);
        ps.setInt(4, credits);
        ps.executeUpdate();
        System.out.println("Course added successfully.");
    }
}
```

```

private static void viewCourses(Connection conn) throws
SQLException {

    String query = "SELECT * FROM courses";

    try (Statement stmt = conn.createStatement(); ResultSet rs =
stmt.executeQuery(query)) {

        while (rs.next()) {

            System.out.printf("ID: %d | Name: %s | Faculty: %s |
Credits: %d%n",

                rs.getInt("course_id"),
                rs.getString("course_name"),
                rs.getString("faculty"),
                rs.getInt("credits"));

        }

    }

}

```

```

private static void updateCourse(Connection conn, Scanner sc)
throws SQLException {

    System.out.print("Enter Course ID to update: ");

    int id = sc.nextInt();

    sc.nextLine();

    System.out.print("Enter new Faculty: ");

    String faculty = sc.nextLine();

    System.out.print("Enter new Credits: ");

    int credits = sc.nextInt();

```

```

        String query = "UPDATE courses SET faculty = ?, credits = ?
WHERE course_id = ?";

        try (PreparedStatement ps = conn.prepareStatement(query)) {
            ps.setString(1, faculty);
            ps.setInt(2, credits);
            ps.setInt(3, id);
            ps.executeUpdate();
            System.out.println("Course updated successfully.");
        }
    }
}

```

```

private static void deleteCourse(Connection conn, Scanner sc)
throws SQLException {
    System.out.print("Enter Course ID to delete: ");
    int id = sc.nextInt();

    String query = "DELETE FROM courses WHERE course_id = ?";
    try (PreparedStatement ps = conn.prepareStatement(query)) {
        ps.setInt(1, id);
        ps.executeUpdate();
        System.out.println("Course deleted successfully.");
    }
}

}
}

```

2.Product Inventory System

```
package JDBCASSES;

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.sql.Statement;
import java.util.Scanner;

public class ProductInventoryApp {
    private static final String URL =
"jdbc:mysql://localhost:3306/testdb";
    private static final String USER = "root";
    private static final String PASSWORD = "root";

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        try (Connection conn = DriverManager.getConnection(URL,
USER, PASSWORD)) {
            while (true) {
                System.out.println("\n--- Product Inventory System ---
");
                System.out.println("1. Add Product");
                System.out.println("2. View Products");
                System.out.println("3. Update Quantity");
                System.out.println("4. Delete Product");
                System.out.println("5. Exit");
                System.out.print("Choose option: ");
```

```

        int choice = scanner.nextInt();

        switch (choice) {
            case 1 -> addProduct(conn, scanner);
            case 2 -> viewProducts(conn);
            case 3 -> updateQuantity(conn, scanner);
            case 4 -> deleteProduct(conn, scanner);
            case 5 -> {
                System.out.println("Exiting...");
                return;
            }
            default -> System.out.println("Invalid option.");
        }
    }
} catch (SQLException e) {
    e.printStackTrace();
}
}

```

```

private static void addProduct(Connection conn, Scanner sc)
throws SQLException {
    System.out.print("Enter Product ID: ");
    int id = sc.nextInt();
    sc.nextLine();
    System.out.print("Enter Product Name: ");
    String name = sc.nextLine();
    System.out.print("Enter Quantity: ");
    int qty = sc.nextInt();
    System.out.print("Enter Price: ");
    double price = sc.nextDouble();

    String query = "INSERT INTO products VALUES (?, ?, ?, ?)";
}

```

```

        try (PreparedStatement ps =
conn.prepareStatement(query)) {
            ps.setInt(1, id);
            ps.setString(2, name);
            ps.setInt(3, qty);
            ps.setDouble(4, price);
            ps.executeUpdate();
            System.out.println("Product added successfully.");
        }
    }
}

```

```

private static void viewProducts(Connection conn) throws
SQLException {
    String query = "SELECT * FROM products";
    try (Statement stmt = conn.createStatement(); ResultSet rs
= stmt.executeQuery(query)) {
        while (rs.next()) {
            System.out.printf("ID: %d | Name: %s | Quantity: %d |
Price: %.2f%n",
                rs.getInt("product_id"),
                rs.getString("product_name"),
                rs.getInt("quantity"),
                rs.getDouble("price"));
        }
    }
}

```

```

private static void updateQuantity(Connection conn, Scanner
sc) throws SQLException {
    System.out.print("Enter Product ID: ");
    int id = sc.nextInt();
    System.out.print("Enter new Quantity: ");
    int qty = sc.nextInt();
}

```



```
String query = "UPDATE products SET quantity = ? WHERE  
product_id = ?";
```

```
try (PreparedStatement ps =  
conn.prepareStatement(query)) {  
    ps.setInt(1, qty);  
    ps.setInt(2, id);  
    ps.executeUpdate();  
    System.out.println("Quantity updated successfully.");  
}  
}
```

```
private static void deleteProduct(Connection conn, Scanner  
sc) throws SQLException {  
    System.out.print("Enter Product ID to delete: ");  
    int id = sc.nextInt();  
    String query = "DELETE FROM products WHERE product_id  
= ?";  
    try (PreparedStatement ps =  
conn.prepareStatement(query)) {  
        ps.setInt(1, id);  
        ps.executeUpdate();  
        System.out.println("Product deleted successfully.");  
    }  
}  
}
```